# Exam

## Table of contents


1 Introduction .......... 1
  1.1 Written exam .......... 1
2 30 points total .......... 1
  2.1 Health data part .......... 1
3 Questions .......... 2
  3.1 1. GDPR .......... 2
  3.2 Grading criteria – Question 2 (2 points) .......... 5
  3.3 2. Reproducibility .......... 5
  3.4 Reproducibility (5 points) .......... 5
  3.5 Grading criteria – Applied workflow question (3 points) .......... 8
  3.6 3. Nomenclature (ICD, ATC etc) .......... 8
  3.7 4. Registers, differnt types of data, quality control, linking .......... 8


# 1 Introduction

Individual written exam, 4 credits (4 hours)

The grading scale comprises:

- Pass with distinction (VG),
- Pass (G) and
- Fail (U).
- The compulsory elements (3.5 credits) are graded Pass (G) and Fail (U),
- the individual written exam (4 credits) is graded Pass with distinction (VG), Pass (G) and Fail (U).

Final grade:

- G: G is required on both the compulsory elements and the written exam.
- VG: G is required on the compulsory elements + VG on the individual written exam.

## 1.1 Written exam

- 18 hp for questionaire part (AGE)
- 12 hp for health data part (EB)
- # 2 30 points total

## 2.1 Health data part

Exluded: - Swedish legislation - Writing Git and SQL code (interpret written code OK)

# 3 Questions

For each: - Yes/no questions concerning definitions etc

## 3.1 1. GDPR

### 3.1.a 1. Yes/no-questions

### (a) Personal data (1.5 points)

For each example below: Is this personal data under GDPR?

1. A dataset containing only ICD-10 codes with no other variables.
2. A dataset containing year of birth, sex, and municipality of residence for individuals in Sweden.
3. A dataset containing only a study ID number and a heading stating "Participants in diabetes study". The key linking the study ID to personal identity numbers is stored separately by another authority and is not accessible to the researcher.
4. A dataset where all identifiers have been permanently removed and re-identification is no longer reasonably possible.
5. A dataset containing Swedish personal identity numbers (PINs) and laboratory results.
6. A dataset containing full names and email addresses of medical doctors who reported data to a register.
7. A dataset containing questionnaire responses about mental health, stress levels, and sleep problems linked to IP addresses that can reasonably be used to identify individuals
8. A dataset containing GPS coordinates from wearable electronic devices that also records heart rate and physical activity.
9. A case report describing "the only 104-year-old woman living in Municipality X" together with a diagnosis history.
10. A dataset containing free-text clinical notes where patient names have been removed, but rare events and detailed circumstances remain.
11. A table with aggregated counts only (e.g., number of cases per region and year), where no cell represents very small counts and individuals cannot reasonably be singled out.
12. A dataset containing pseudonymised IDs only, where the key has been permanently destroyed and re-identification is no longer reasonably possible.

Solution

**Answers – Personal data** 1. **No.** ICD-10 codes alone do not relate to an identified or identifiable person.

2. **Yes.** A combination of demographic variables may make a person indirectly identifiable.
3. **Yes.** The data are pseudonymised, not anonymised. The existence of a separate key means that individuals remain identifiable in principle, even if the researcher does not have access to the key.
4. **No.** Truly anonymised data fall outside the scope of GDPR.
5. **Yes.** A PIN is a direct identifier; this clearly constitutes personal data.
6. **Yes.** Names and email addresses relate to identifiable natural persons, even if in a professional role.
7. **Yes.** Questionnaire responses linked to IP addresses that can reasonably identify individuals constitute personal data.
8. **Yes.** Location data and device-generated data linked to individuals can make a person identifiable, directly or indirectly.
9. **Yes.** Even without a name, this description is likely to single out an identifiable individual (indirect identification).
10. **Yes.** Free text may allow indirect identification even after obvious identifiers are removed.
11. **No.** Properly aggregated statistics that do not allow identification are not personal data.
12. **No.** If the key has been permanently destroyed and re-identification is no longer reasonably possible, the data are anonymised and fall outside GDPR.

### 3.1.b (b) Data concerning health (1 point)

From the examples above, list the numbers of those that qualify as **data concerning health** under GDPR. (Only list the numbers.)

**(c) Processing (1.5 points)**

### 3.1.c (c) Processing (6 points)

For each example below: Does this constitute **processing under GDPR**?

1. Storing personal data on a secure university server.
2. Deleting personal data.
3. Receiving personal data from another authority.
4. Viewing personal data without modifying them.
5. Encrypting personal data.
6. Replacing personal identity numbers with study IDs (pseudonymisation).
7. Transferring personal data to another EU university for research.
8. Running a statistical analysis on anonymised data where re-identification is no longer reasonably possible.
9. Destroying a paper printout containing personal data.

10. Writing a research article that reports only fully aggregated results where no individuals can reasonably be identified.
11. Backing up a dataset containing personal data.
12. Analysing synthetic data that do not relate to any real individuals.

Solution

**Answers:**

1. Yes
2. Yes
3. Yes
4. Yes
5. Yes
6. Yes
7. Yes
8. No
9. Yes
10. No
11. Yes
12. No

Processing includes any operation performed on personal data, such as collection, storage, consultation, use, disclosure, restriction, erasure, or destruction.

However, operations performed on truly anonymised or synthetic data that do not relate to identifiable individuals do not constitute processing under GDPR.

### 3.1.d Essay (2 points)

You are a statistician at a university. Your supervisor gives you a dataset containing health data and asks you to start analysing it for a research project. Before beginning the analysis, you want to ensure that the processing is lawful and compliant with GDPR. List and briefly explain at least four questions you should clarify before starting the work.

**Important:** Only legal and governance aspects under GDPR are relevant for this question. Points may be deducted if the answer includes clearly incorrect legal statements. Questions related to statistical methodology, study design, research hypotheses, practical deadlines, or other administrative matters are not relevant in this context and will not be awarded points.

Solution

## 3.2 Grading criteria – Question 2 (2 points)

**Maximum: 2 points**

Points are awarded based on the number of relevant and correctly explained GDPR-related aspects identified.

#### 3.2.a Full score (2 points)

- The answer identifies **at least four relevant and clearly explained GDPR-related aspects**, such as:
  - Identification of the **data controller and data processor**
  - The **lawful basis** for processing (Article 6 and, where applicable, Article 9)
  - Whether the dataset contains **special categories of personal data** (e.g. health data)
  - The existence of appropriate **technical and organisational safeguards**
  - **Purpose limitation**
  - **Data minimisation**
  - Existence of a **Data Processing Agreement (DPA)** where applicable
  - Retention period considerations
- Concepts are used correctly and demonstrate structured understanding.

#### 3.2.b Partial credit (1–1.5 points)

- The answer identifies **two or three relevant aspects**.
- Some explanations may be brief or partially developed.
- Minor inaccuracies may be present but do not reflect fundamental misunderstandings.

#### 3.2.c Low credit (0–0.5 points)

- Fewer than two relevant aspects are identified.
- The answer is vague or lacks connection to GDPR concepts.
- Major misunderstandings of core concepts (e.g. pseudonymisation vs anonymisation, controller vs processor).

## 3.3 2. Reproducibility

## 3.4 Reproducibility (5 points)

#### 3.4.a Part A – Short statements (2 points)

For each statement below, answer **Yes or No**.
(0.2 points per correct answer)

1. If a study shares its raw data but not the analysis code, the study is fully reproducible.
2. Reproducibility and replicability mean the same thing.
3. If all analysis steps were performed manually in a spreadsheet but the final results are reported correctly, the study is reproducible.

4. Making code available automatically guarantees that the scientific conclusion is correct.
5. A computational pipeline (e.g., using `{targets}`) automatically makes an analysis reproducible.
6. Using version control (e.g., Git/GitHub) improves reproducibility because it allows researchers to track changes in code over time.
7. If independent researchers cannot access the original data due to legal restrictions, the study can still be reproducible in principle.
8. Reproducibility is primarily a statistical issue rather than a computational one.
9. Documenting software versions and package dependencies is relevant for reproducibility.
10. A study can be reproducible even if the results cannot be replicated in a new dataset.

Solution

**Answers and brief justification**

1. No. Reproducibility requires both analytic data and the code used to generate the results (Peng & Hicks).
2. No. Reproducibility concerns the same data and code; replicability concerns new data (Peng & Hicks).
3. No. Undocumented manual steps prevent others from recreating the workflow.
4. No. Reproducibility does not guarantee validity (Peng & Hicks; Baker).
5. No. A pipeline facilitates reproducibility but does not automatically guarantee it.
6. Yes. Version control improves traceability of analytical changes.
7. Yes. Reproducibility is a property of the analysis; legal access restrictions affect feasibility, not the definition itself.
8. No. Reproducibility is strongly linked to computational transparency and workflow documentation.
9. Yes. Differences in software versions may affect results and prevent exact reproduction.
10. Yes. A result can be reproducible but fail to replicate in new data (Peng & Hicks; reproducibility ≠ replicability).

### 3.4.b Part B – Applied workflow question (3 points)

You are hired as a statistician to update the annual report of a national quality register.

You receive:

- Several R scripts that must be run in a specific order.
- Manual instructions describing which scripts to run first.
- Intermediate datasets saved with names such as "final2_new_version_revised.csv".
- No version control.
- No documentation of software versions.
- The previous report was produced last year and next year's report should contain the same analyses, but for a new time period.

You are asked to ensure that future annual reports can be produced in a transparent and reproducible way.

Describe how you would restructure the workflow and the code.
Your answer should:

- Explain which tools or principles you would introduce.
- Briefly explain why these changes improve reproducibility.

Solution

**Example answer elements**

Proposed restructuring:

1. Implement a computational pipeline (e.g., `{targets}`):
   - Explicitly define dependencies between data import, cleaning, modelling, and reporting.
   - Ensure that only affected steps rerun when data change.
   - Avoid manual script ordering.
2. Introduce version control (e.g., Git):
   - Track changes in code over time.
   - Tag the version used for each annual report.
   - Enable rollback to earlier states.
3. Separate raw data, derived data, and output:
   - Never overwrite raw data.
   - Keep outputs reproducible from source code.
4. Document the computational environment:
   - Record R and package versions.
   - Ensure that the same setup can be recreated in future years.

These changes improve transparency, traceability, and ensure that each annual report can be regenerated using the same structured workflow.

### 3.5 Grading criteria – Applied workflow question (3 points)

Full credit (3 points):

- Clearly identifies at least three reproducibility risks in the current workflow.
- Explicitly proposes structured improvements (e.g., pipeline framework, version control).
- Explains why these tools improve reproducibility (dependency tracking, traceability, automation, version history).

Partial credit (1.5–2.5 points):

- Identifies some risks but explanations are incomplete.
- Mentions tools but does not clearly explain their function.

Low credit (0–1 point):

- Focuses mainly on statistical method quality rather than workflow reproducibility.
- Provides vague suggestions (e.g., "organize the code better") without linking to specific tools or principles.

## 3.6 3. Nomenclature (ICD, ATC etc)

## 3.7 4. Registers, different types of data, quality control, linking